225760: tuffaceous volcanic rock, Wittenoom Gorge

(Dales Gorge Member, Brockman Iron Formation, Hamersley Group, Hamersley Basin, Capricorn Orogen)

Location and sampling

Mount Bruce (SF 50-11), Wittenoom (2553)
MGA Zone 50, 636783E 7530427N

Sampled on 12 September 2017

This sample was collected from outcrop on the east side of a pool on Joffre Creek in Wittenoom Gorge, about 23 km east-southeast of Mount King, 9.2 km north-northeast of Joffre Falls, and 1.4 km southwest of Club Pool.

Geological context

The unit sampled is a tuffaceous volcanic rock (Fig. 1) within the 2494–2454 Ma Brockman Iron Formation of the Hamersley Group (Martin, 2020a). The Brockman Iron Formation is about 0.5 km thick, consists of banded iron-formation, chert, mudstone, and siltstone, with minor dolerite sills, and conformably overlies the Mount McRae Shale and is conformably overlain by the Weeli Wolli Formation (Martin, 2020a). Numerous tuffaceous horizons within the Brockman Iron Formation have been dated, and indicate depositional ages between 2495 and 2454 Ma (Trendall et al., 1998; Pickard, 2002; Trendall et al., 2004). Tuffaceous horizons within the Dales Gorge Member are marker beds that are commonly referred to as shale bands (Trendall and Blockley, 1968) and abbreviated as DS1–DS16. The volcanic rock sampled at this locality is from DS2 near the base of the 2494–2464 Ma Dales Gorge Member, which is 120–180 m thick and the lowermost of four members within the Brockman Iron Formation (Martin, 2020b). An underlying tuffaceous volcanic rock (DS1) from this locality yielded an igneous crystallization age of 2485 ± 9 Ma (GSWA 225758, Wingate et al., 2021c). Volcaniclastic rocks within DS9, sampled about 0.9 km to the southwest, yielded maximum depositional ages of 2477 ± 2 Ma (GSWA 225756, Wingate et al., 2021a) and 2477 ± 4 Ma (GSWA 225757, Wingate et al., 2021b).

Petrographic description

The sample is an altered tuffaceous volcanic rock, consisting of >97% clay minerals, and trace chlorite, hematite, and quartz. Clay minerals (probably smectite) are cryptocrystalline to very fine grained and moderately birefringent. The rock contains 10–15% poorly defined colourless, oval to elongate patches, which are up to 0.6 mm long. The patches include fine-grained clay particles, although the colourless parts have negligible birefringence and no crystal form is visible even at high magnification. Their composition is therefore unclear, but they could be cryptocrystalline kaolinite or another kaolin group mineral. Traces of very pale green chlorite form diffuse patches up to 0.1 mm long. Disseminated subhedral to euhedral hematite and extremely rare angular quartz grains are up to 0.05 mm in size. There is no indication of deformation or metamorphism.

Zircon morphology

Twenty-four zircons isolated from this sample are colourless to pale brown and anhedral to subhedral. The crystals are up to 100 µm long, and equant to slightly elongate, with aspect ratios up to 3:1. In cathodoluminescence (CL) images, most zircons exhibit concentric zoning. A CL image of all zircons is shown in Figure 2.

Analytical details

This sample was analysed on 18–19 February 2019, using SHRIMP-B. Fifteen analyses of the M257 standard obtained during the session indicated an external spot-to-spot (reproducibility) uncertainty of 0.50% (1σ) and a $^{238}U/^{206}Pb^{*}$ calibration uncertainty of 0.12% (1σ). Isotopic mass fractionation of $^{207}Pb/^{206}Pb$ ratios during the session was corrected by reference to the OGC1 standard; measured ratios were decreased by 0.13%. Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^{*}$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Sixteen analyses were obtained from 16 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

Interpretation

The analyses are concordant to moderately discordant (Fig. 3). Two analyses are >5% discordant. The dates obtained from these two analyses (Group D; Table 1) are unreliable, and are considered not to be geologically significant. The remaining 14 analyses can be divided into two groups, based on their $^{207}Pb^{*}/^{206}Pb^{*}$ ratios.

Figure 1. Outcrop image for sample 225760: tuffaceous volcanic rock, Wittenoom Gorge. The layer sampled for geochronology is indicated

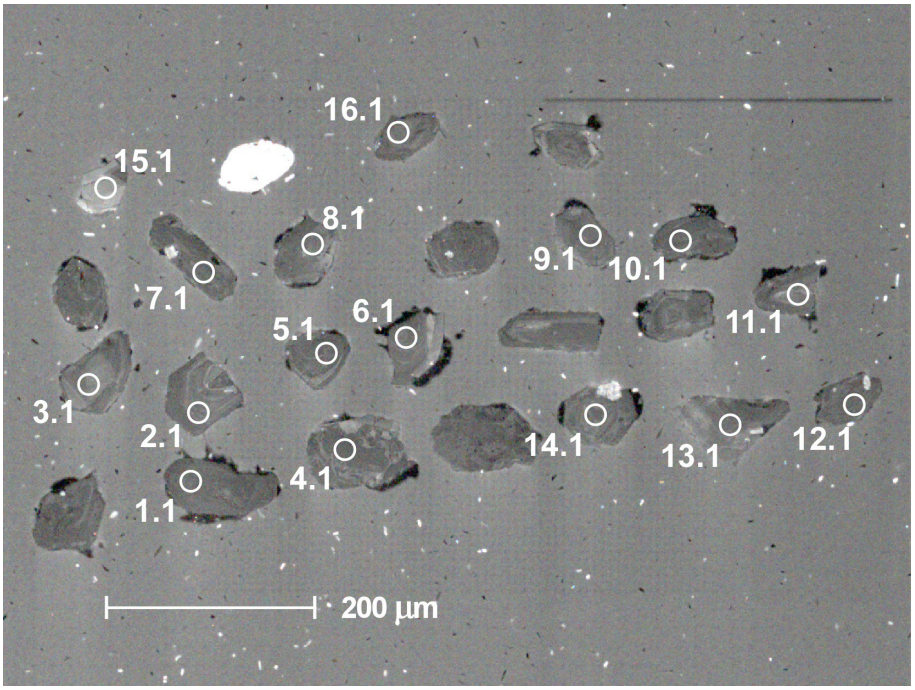

Figure 2. Cathodoluminescence image of all zircons from sample 225760: tuffaceous volcanic rock, Wittenoom Gorge. Numbered circles indicate the approximate locations of analysis sites

Table 1. Ion microprobe analytical results for zircons from sample 225760: tuffaceous volcanic rock, Wittenoom Gorge

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
I	3	3.1	169	137	0.81	0.059	2.157	0.019	0.16119	0.00088	2.159	0.019	0.16066	0.00090	2454	19	2463	9	0.4
I	5	5.1	385	1004	2.61	0.096	2.169	0.016	0.16166	0.00066	2.171	0.016	0.16081	0.00068	2443	15	2464	7	0.9
I	15	15.1	40	45	1.15	0.115	2.150	0.032	0.16331	0.00158	2.152	0.032	0.16229	0.00169	2460	31	2480	18	0.8
I	6	6.1	152	193	1.27	-0.012	2.150	0.021	0.16231	0.00095	2.150	0.021	0.16241	0.00096	2462	20	2481	10	0.8
I	4	4.1	510	760	1.49	0.021	2.186	0.015	0.16263	0.00062	2.187	0.015	0.16244	0.00063	2428	14	2481	7	2.2
I	13	13.1	229	158	0.69	0.030	2.195	0.018	0.16285	0.00085	2.196	0.018	0.16258	0.00086	2419	17	2483	9	2.5
I	7	7.1	244	119	0.49	0.129	2.214	0.018	0.16382	0.00125	2.217	0.018	0.16267	0.00128	2400	16	2484	13	3.4
I	2	2.1	165	153	0.93	-0.021	2.167	0.020	0.16266	0.00091	2.166	0.020	0.16285	0.00092	2447	19	2485	9	1.6
I	1	1.1	312	83	0.27	0.046	2.168	0.017	0.16330	0.00073	2.169	0.017	0.16289	0.00074	2444	16	2486	8	1.7
I	9	9.1	189	270	1.43	0.082	2.133	0.019	0.16372	0.00085	2.134	0.019	0.16299	0.00089	2477	18	2487	9	0.4
I	8	8.1	147	71	0.49	-0.012	2.185	0.021	0.16379	0.00095	2.184	0.021	0.16389	0.00095	2430	19	2496	10	2.7
X	16	16.1	105	122	1.16	0.013	2.013	0.021	0.17934	0.00104	2.013	0.021	0.17922	0.00105	2600	22	2646	10	1.7
X	11	11.1	197	156	0.79	0.250	1.990	0.017	0.18181	0.00086	1.995	0.017	0.17959	0.00095	2619	18	2649	9	1.1
X	10	10.1	352	238	0.68	0.000	1.965	0.016	0.18236	0.00070	1.965	0.016	0.18236	0.00070	2653	17	2674	6	0.8
D	12	12.1	512	353	0.69	0.136	2.513	0.017	0.16029	0.00064	2.516	0.017	0.15908	0.00068	2157	13	2446	7	11.8
D	14	14.1	556	353	0.63	0.125	2.295	0.015	0.16296	0.00060	2.298	0.015	0.16184	0.00064	2329	13	2475	7	5.9

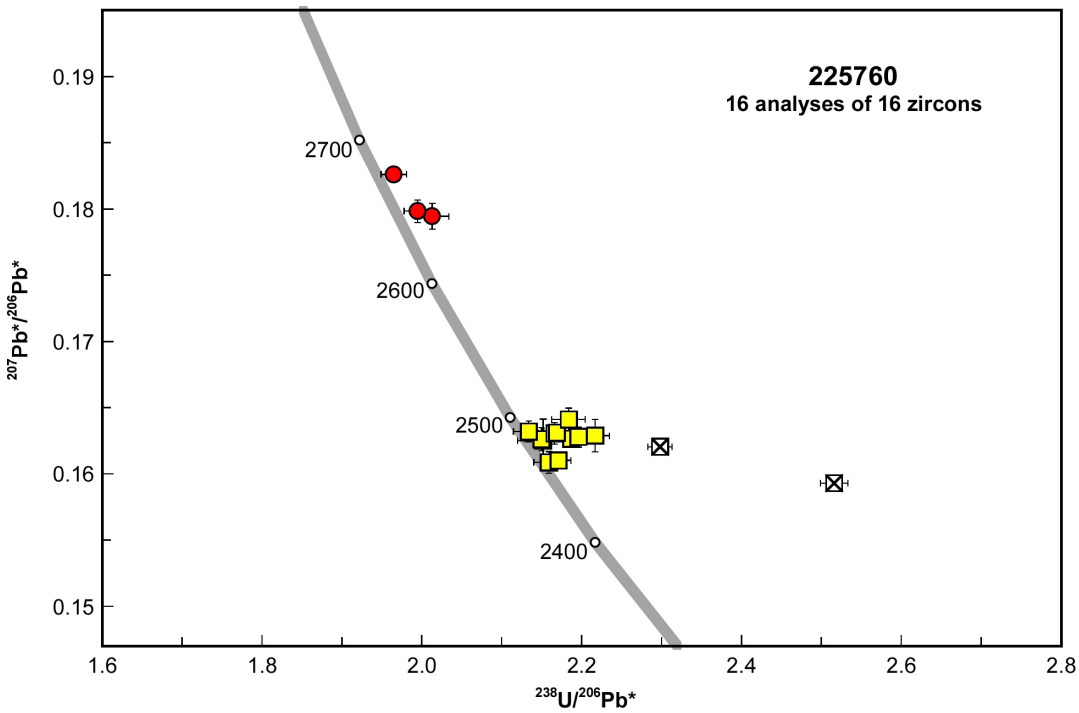

Figure 3. U–Pb analytical data for zircons from sample 225760: tuffaceous volcanic rock, Wittenoom Gorge. Yellow squares indicate Group I (magmatic zircons); red circles indicate Group X (xenocrystic zircons); crossed squares indicate Group D (discordance >5%)

Group I comprises 11 analyses (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2480 ± 7 Ma (MSWD = 1.3).

Group X comprises three analyses (Table 1), which yield $^{207}Pb^*/^{206}Pb^*$ dates of 2674–2646 Ma.

The date of 2480 ± 7 Ma for the 11 analyses in Group I is interpreted as the igneous crystallization age. The dates of 2674–2646 Ma for the three analyses in Group X are interpreted as the ages of inherited zircons.

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Recommended reference for this publication

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